



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
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PROJECT TITLE

A CAPSTONE PROJECT REPORT

Improving Organ Donar-Recipient Matching Using Graph Algorithms: A Smart Organ Donation System Compared with Traditional Matching Methods

CSA0305-DATA STRUCTURE

to the award of the degree of

BACHELOR OF TECHNOLOGY IN

Artificial Intelligence

Submitted by

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Under the Supervision of

Guide Name

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Improving Organ – Donor Recipient Matching using Graph Algorithm A Smart Organ Donation System Compared with Traditional Methods**” has been carried out by **Ragul.m (192524066)** under the supervision of **Dr.Uma priyadarsini** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, **Sree B, Trivikram N.S.V, Siva Charan O, Ragul M, and Dhanush** of the **Computer Science and Engineering Department**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**Improving Organ – Donor Recipient Matching using Graph Algorithm A Smart Organ Donation System Compared with Traditional Methods**” is the result of our own analyses efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Sree B (192524023)	Donor & Recipient Registration and Medical Profile Management	Developed registration, profiles and matching	Tested registration and matching	Modules 1	<u>20</u> %
Trivikram (192525063)	Compatibility Analysis and Graph-Based Match Scoring	Developed compatibility and match scoring	Tested matching accuracy	Modules 2	20%
O Siva Charran (192525039)	Requests & Allocation	Developed requests, approval and priority allocation	Tested workflows and allocation	Modules 3	20%
Ragul M (192524066)	Dashboard & Reports	Developed dashboard, analytics and reports	Performed integration and system testing	Modules 4	20%
R Dhanush (192525318)	UI & System Integration	Developed dashboard, analytics and reports	Performed integration and system testing	Modules 4	20%
Total					100%

ABSTRACT

Organ transplantation is a critical medical procedure that provides a life-saving treatment for patients suffering from severe organ failure. The success of transplantation largely depends on identifying a suitable donor for a recipient within a limited period. Organ matching involves several important factors, including blood group compatibility, organ type, tissue compatibility, medical urgency, waiting time, and recipient priority. Traditional organ matching systems generally use predefined rules and sequential searching methods to identify compatible recipients. As the number of donors and recipients increases, these methods can become time-consuming and may not efficiently identify the most suitable donor-recipient combination. Therefore, there is a need for an intelligent and efficient approach that can analyse multiple matching factors simultaneously.

This project, **“Improving Organ Donor-Recipient Matching Using Graph Algorithms: A Smart Organ Donation System Compared with Traditional Matching Methods,”** proposes a graphbased approach for improving the organ donation matching process. In the proposed system, donors and recipients are represented as nodes in a graph, while compatibility relationships between them are represented as edges. Each donor-recipient connection can be evaluated using multiple factors to determine the level of compatibility. Graph algorithms can then be applied to analyze these relationships, identify suitable matches, and prioritize recipients according to their compatibility and medical requirements. This representation allows the system to handle multiple possible donorrecipient relationships in an organized and efficient manner.

Keywords

Organ Transplantation, Organ Donation, Donor-Recipient Matching, Graph Algorithms, Smart Organ Donation System, Compatibility Analysis, Medical Urgency, Recipient Priority, Blood Group Compatibility, Organ Allocation.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
D	Number of Donors	Count
R	Number of Recipients	Count
M	Number of Compatible Matches	Count
U	Recipient Urgency Level	Scale (1–10)
B	Blood Group Compatibility Score	Percentage (%)
O	Organ Compatibility Score	Percentage (%)
L	Location Compatibility Score	Percentage (%)
S	Overall Match Score	Percentage (%)
T	Matching Response Time	Seconds
P	Priority Score	Numerical Score

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Organ transplantation is an important medical procedure that can save the lives of patients suffering from serious organ failure. The success of transplantation depends on finding a suitable donor-recipient match based on factors such as blood group, organ type, tissue compatibility, medical urgency, and waiting time. Traditional organ matching systems generally use predefined rules and sequential searching methods to identify compatible recipients. As the number of donors and recipients increases, these methods may require more processing time and may not efficiently identify the most suitable match.

1.2 Need for the Project

Why the Problem Needs to Be Addressed

Finding suitable organ donors quickly is essential to save patients' lives.

Manual and traditional matching methods can be slow and inefficient.

Multiple medical factors must be considered simultaneously.

Who Is Affected by the Problem

Patients waiting for organ transplantation.

Organ donors and their families.

Hospitals and healthcare professionals.

Organ donation and transplantation organizations.

Existing Limitations

Traditional systems use predefined rules and sequential searching.

Difficulty in handling multiple matching factors efficiently.

Slow identification of the most suitable donor-recipient pair.

Limited optimization of recipient priority.

Engineering Significance

Graph algorithms enable efficient analysis of donor-recipient relationships.

The system can process multiple compatibility factors.

Improves matching accuracy and organ allocation.

Reduces processing time and supports better medical decision-making.

Problem Statement

Provide a precise engineering problem statement.

The problem should, wherever appropriate, involve:

- Multiple interacting factors
- Technical complexity
- Conflicting requirements
- Realistic engineering constraints

1.3 Project Aim

- To develop a smart organ donation and recipient matching system.
- To use graph algorithms for efficient donor-recipient matching.
- To improve compatibility analysis and recipient prioritization.
- To reduce the time required for organ allocation.

1.4 Project Objectives

- To design a database for managing donor and recipient information.
- To develop a smart organ donation and recipient registration system.
- To model donor-recipient relationships using graph structures.
- To implement graph algorithms for organ matching and priority allocation.
- To test the system using different compatibility and medical urgency factors.
- To evaluate the performance of the proposed system compared with traditional matching methods....

1.5 Scope

What is Included

- Donor and recipient registration.
- Medical profile and organ information management.
- Graph-based donor-recipient matching.
- Compatibility and priority analysis.
- Organ request, approval, and status tracking.

- Admin monitoring and report generation.

What is Excluded

- Actual medical transplantation procedures.
- Real-time hospital equipment integration.
- Online financial or insurance processing.
- Direct replacement of doctors' medical decisions.

Assumptions

- Donor and recipient medical information is accurate.
- Blood group and organ details are available.
- Authorized users provide valid information.
- Medical professionals verify final matches.

Boundary Conditions

- The system operates using registered donor and recipient data.
- Matching is limited to the supported organs and compatibility criteria.
- Final organ allocation requires medical and administrative approval.
- System performance depends on the availability of accurate data.

1.6 Expected Engineering Outcome

What is Included

- Donor and recipient registration.
- Medical and organ information management.
- Graph-based organ matching.
- Compatibility and priority analysis.
- Request and status management.
- Admin monitoring and reporting.

What is Excluded

- Actual transplantation procedures.
- Real-time hospital equipment integration.
- Financial and insurance processing.
- Replacement of medical professionals' decisions.

Assumptions

- Medical information is accurate.
- Required donor and recipient details are available.
- Users provide valid information.
- Final matches are verified by medical professionals.

Boundary Conditions

- Only registered users are considered.
- Matching depends on supported organs and compatibility factors.
- Final allocation requires approval.
- Results depend on the accuracy of available data.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature and existing solutions were identified through:

- Research papers on organ transplantation and matching systems.
- Studies related to graph algorithms and healthcare applications.
- Existing organ donation and allocation systems.
- Medical compatibility and recipient-priority methods.
- Online databases, journals, and technical publications.

2.2 Review of Existing Work

Research Papers

- Existing studies use algorithms to improve donor-recipient matching.
- Machine learning and graph-based techniques help analyze multiple matching factors.
- Research focuses on improving matching accuracy and allocation efficiency.

Existing Systems

- Traditional systems use predefined rules for organ allocation.
- Donors and recipients are matched based on medical compatibility and priority.
- Manual or sequential searching can become inefficient for large datasets.

Commercial Solutions

- Healthcare and organ transplant organizations use centralized databases.
- These systems manage donor and recipient information.
- Most systems rely on predefined medical allocation policies.

Technologies

- Databases are used to store medical and patient information.
- Web technologies provide access to organ donation services.
- Graph data structures can represent relationships between donors and recipients.

Methods

- Rule-based matching is commonly used in traditional systems.
- Compatibility scoring evaluates factors such as blood group and organ type.
- Graph algorithms can identify and prioritize suitable matches efficiently.

Research Gap

- Traditional methods may not efficiently handle multiple donor-recipient relationships.
- Limited optimization can affect the selection of the most suitable recipient.
- A graph-based approach can improve relationship analysis, priority scoring, and organ allocation.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Traditional RuleBased Matching	Simple and easy to implement	Limited optimization and slow for large data	Improved using graphbased matching
Manual Organ Allocation Systems	Allows medical expert decisions	Time-consuming and prone to delays	Automation improves allocation efficiency
Priority-Based Matching	Considers medical urgency	May not analyze all compatibility factors	Enhanced using multiple priority factors
Database-Based Matching Systems	Efficient data storage and management	Limited relationship analysis	Graph models improve donor-recipient relationships
Graph-Based Matching Methods	Efficiently analyses multiple relationships	Requires algorithm design and quality data	Forms the core approach of the proposed system

Table 1: Comparison of Existing Organ Donation Matching Approaches

2.4 Research / Engineering Gap

- Traditional matching methods mainly use predefined rules and sequential searching.
- Multiple donor-recipient relationships are difficult to analyze efficiently.
- Existing systems may not optimize compatibility, urgency, and priority together.
- Large amounts of donor and recipient data can increase matching complexity.
- There is a need for an intelligent graph-based approach for efficient organ allocation.

2.5 Proposed Contribution

- Represents donors and recipients as nodes in a graph.
- Uses compatibility relationships as graph edges.
- Analyzes multiple factors, including blood group, organ type, urgency, and priority.
- Applies graph algorithms to identify suitable matches efficiently.
- Provides compatibility scoring and priority-based organ allocation.
- Improves matching efficiency compared with traditional sequential approaches.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder	Requirement
Donor	Easy registration and secure profile management
Recipient	Ability to submit organ requests and view matching status
Medical Professional	Access to compatibility and priority information
Administrator	Monitor users, matches, requests, and system reports

Table 2: Stakeholder Requirements Analysis

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
User registration	Functional	Successful registration of valid users
Organ matching	Functional	Match based on defined compatibility factors
Priority calculation	Functional	Generate a priority score for recipients
Request management	Functional	Track request status accurately
Response time	Non-Functional	Results generated within a few seconds
Data security	Non-Functional	Authorized access to sensitive data
Usability	Non-Functional	Simple and user-friendly interface

Table 3: Functional and Non-Functional Requirements of the Proposed System

3.2 Constraints & Applicable Standards

Constraints

- Availability and accuracy of medical data.
- Limited access to real hospital databases.
- The system supports decision-making and does not replace medical professionals.
- Development is limited by available time and computing resources.
- Sensitive personal and medical information requires secure handling.

Applicable Standards

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 27001	Information security management	Applied access control and secure data handling
OWASP Guidelines	Secure web application development	Used input validation and authentication
Healthcare Privacy Principles	Protection of medical information	Restricted unauthorized access to user data

Table 4: Standards, Security Codes, and Privacy Guidelines Applied in the Proposed System

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Matching Accuracy	High compatibility identification	%	High
Response Time	Fast match generation	Seconds	High
User Capacity	Support multiple users	Users	Medium
Data Security	Authorized access	Boolean	High
Availability	System accessible when required	%	High

Table 5: Performance and Quality Requirements of the Proposed System

3.4 Alternative Solutions & Evaluation

Alternative 1: Traditional Rule-Based Matching

This approach uses predefined rules to match donors and recipients. It is simple but becomes less efficient when multiple matching factors are involved.

Alternative 2: Machine Learning-Based Matching

This approach uses historical data to predict suitable matches. However, it requires large, high-quality datasets and increases system complexity.

Alternative 3: Graph-Based Matching

This approach represents donors and recipients as nodes and their compatibility as relationships. It efficiently analyzes multiple possible matches and supports priority-based allocation

Criterion	Weight	Rule1 Alternative Based	ML 2- Alternative Based	Alternative 3 Graph- Based
Performance	30%	5	4	5
Cost	15%	4	2	4
Safety	20%	4	3	5
Sustainability	10%	3	3	4
Reliability	25%		4	5

Table 6: Weighted Decision Matrix for Selection of Organ Matching Approach

Selected Approach: Graph-Based Matching

3.5 Selected Design & Detailed Design

The graph-based approach was selected because it achieved the best overall evaluation in terms of performance, safety, and reliability. It can efficiently represent and analyze multiple donor/recipient relationships.

System Flow



Figure 1 System Flow

The registration module collects donor and recipient information and stores it in the database. The matching module retrieves this information and creates compatibility relationships. The priority module ranks suitable recipients, while the request management module handles approval and status updates. Finally, the administrator monitors the overall system and generates reports.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Graph-based matching	Rule-based matching	Handles multiple relationships efficiently	Higher implementation complexity	Graph approach selected for better matching
Priority scoring	First-comefirst-served	Considers medical urgency	Requires multiple calculations	Priority scoring selected for fair allocation
Web-based system	Desktop application	Easy accessibility	Requires network access	Web system selected for wider access
Automated matching	Manual matching	Faster processing	Requires accurate data	Automated matching selected for efficiency

Table 7: Design Decisions and Justification for the Proposed System

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Efficient use of computing resources	Software-based system reduces paper records	Digital system selected
Ethics	Fair organ allocation	Priority and compatibility factors	Transparent scoring approach used
Health & Safety	Incorrect medical matching	Medical factors and professional verification	System supports, not replaces, doctors
Security	Protection of medical data	Risk of unauthorized access	Authentication and access control included

Table 8: Broader Considerations and Their Impact on System Design

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect user data	Medium	High	High	Data validation and verification	Medium
Unauthorized access	Medium	High	High	Authentication and access control	Low
Incorrect matching	Low	High	High	Medical verification of results	Medium
System failure	Low	Medium	Medium	Database backup and error handling	Low
Data loss	Low	High	Medium	Regular data backup	Low

Table 9: Risk Assessment and Mitigation Plan for the Proposed System

CHAPTER 4

ORGAN REQUEST, APPROVAL and STATUS MANAGEMENT

4.1 Development Approach & Resources

The project was developed using an iterative approach. The system was designed module by module, including registration, profile management, organ matching, priority scoring, request management, and administration. Each module was implemented and tested before integration into the complete system.

Resources Used: Donor-recipient sample data, medical compatibility criteria, graph algorithms, Java, JSP, Servlets, MySQL, Eclipse IDE, and Apache Tomcat.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

The project was implemented as a web-based application. Java Servlets were used for backend processing, JSP was used for the user interface, and MySQL was used for database management. Graph-based algorithms were implemented to analyze donor-recipient compatibility and prioritize suitable matches.

Tool / Software	Purpose	Stage Used
Java	Backend development	Implementation
JSP	User interface development	Implementation
Java Servlets	Request processing	Implementation
MySQL	Database management	Development
Eclipse IDE	Code development	Development
Apache Tomcat	Application deployment	Testing
Graph Algorithms	Donor-recipient matching	Core implementation

Table 10: Tools and Technologies Used for System Development

Implementation Evidence: Include screenshots of the registration page, matching results, request status page, admin dashboard, and database.

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
User Registration	Enter valid user details	User registered successfully
Login	Enter valid credentials	User accesses the system
Organ Matching	Provide compatible donorrecipient data	Suitable matches displayed
Priority Scoring	Test different urgency levels	Correct priority score generated
Request Management	Send and approve requests	Status updated correctly
Admin Monitoring	Access dashboard	User and matching data displayed

Table 11: System Testing and Acceptance Criteria

Verification Results

Specification	Required Value	Achieved Value	Status
Registration	Successful	Successful	Pass
Login	Successful	Successful	Pass
Compatibility Matching	Correct match	Correct match	Pass
Priority Calculation	Accurate result	Accurate result	Pass
Request Status	Correct update	Correct update	Pass
Admin Dashboard	Data displayed	Data displayed	Pass

Table 12: Verification of System Specifications and Achieved Result

4.4 Results

The developed system successfully performed the following functions:

- Donor and recipient registration.
- Medical profile management.
- Compatibility-based organ matching.
- Graph-based relationship analysis.
- Match score calculation.
- Priority-based recipient allocation.
- Organ request and approval management.
- Request status tracking.
- Administrative monitoring and reporting.

Results Evidence: Insert screenshots of the implemented modules and matching results.

4.5 Data Analysis & Engineering Judgment

The system was tested using different donor and recipient records. The graph-based approach successfully identified multiple possible relationships between donors and recipients.

The compatibility score was calculated using factors such as:

- Blood group compatibility.
- Organ type.
- Medical urgency.
- Recipient priority.
- Location and other available criteria.

The results showed that the proposed approach can organize and analyze multiple donor-recipient relationships more effectively than a simple sequential matching approach. However, the accuracy of the final result depends on the quality and completeness of the available medical data.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design a donor-recipient database	Database tables implemented	Fully Achieved
Develop a registration system	Registration modules working	Fully Achieved
Model relationships using graphs	Donor-recipient relationships analyzed	Fully Achieved
Implement priority-based matching	Match scores generated	Fully Achieved
Test the system	Functional modules tested	Fully Achieved
Evaluate the proposed approach	Results compared with traditional matching	Partially Achieved

Table 13: Evaluation of Project Objectives and Achievement Levels

4.7 Strengths & Limitations

Objective	Evidence	Achievement
Design a donor-recipient database	Database tables implemented	Fully Achieved
Develop a registration system	Registration modules working	Fully Achieved
Model relationships using graphs	Donor-recipient relationships analyzed	Fully Achieved
Implement priority-based matching	Match scores generated	Fully Achieved
Test the system	Functional modules tested	Fully Achieved
Evaluate the proposed approach	Results compared with traditional matching	Partially Achieved

Table 14: Project Objectives, Evidence and Achievement Evaluation

4.7 Strengths & Limitations

Strengths

- Efficient donor-recipient data management.
- Graph-based representation of relationships.
- Multiple compatibility factors considered.
- Priority-based matching and allocation.
- Easy-to-use web interface.
- Request and status tracking.

Limitations

- Uses sample or manually entered medical data.
- Does not connect directly with hospital databases.
- Final medical decisions require professional verification.
- Advanced medical factors may not be included.
- Performance may vary with very large datasets.

4.8 Project Planning, Roles & Budget

Project Milestones

Phase	Activity
Phase 1	Problem identification and requirement analysis
Phase 2	System and database design
Phase 3	Module development
Phase 4	Graph algorithm implementation
Phase 5	Testing and debugging
Phase 6	System integration
Phase 7	Documentation and final presentation

Table 15: Project Development Phases and Activities

Student Roles & Contributions

Student	Assigned Responsibility	Actual Contribution
Student 1	Registration and graph-based matching	20%
Student 2	Compatibility analysis and match scoring	20%
Student 3	Request, approval	20%
Student 4	Admin dashboard, testing and reporting	20%
Student 5	Organ Matching and Approval	20%

Table 16: Team Member Responsibilities and Contribution Distribution

Project Budget

Item	Estimated Cost	Actual Cost
Software Development Tools	₹0	₹0
Eclipse IDE	₹0	₹0
Apache Tomcat	₹0	₹0
MySQL Database	₹0	₹0
Internet and Testing	₹500	₹500
Documentation and Printing	₹500	₹500
Total	₹1,000	₹1,000

Table 17: Project Cost Estimation and Actual Expenditure



Figure 1: Organ Requests Page



Figure 2: Screenshot of Requests page

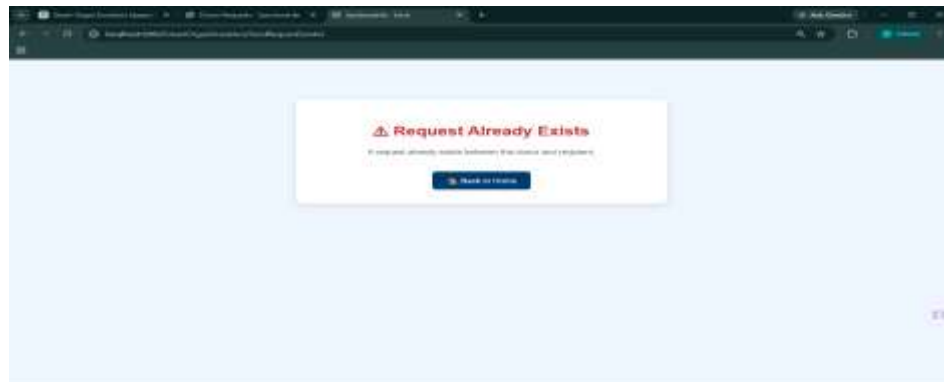


Figure 3: screenshot of prevents conflicting multiple acceptances for the same recipient

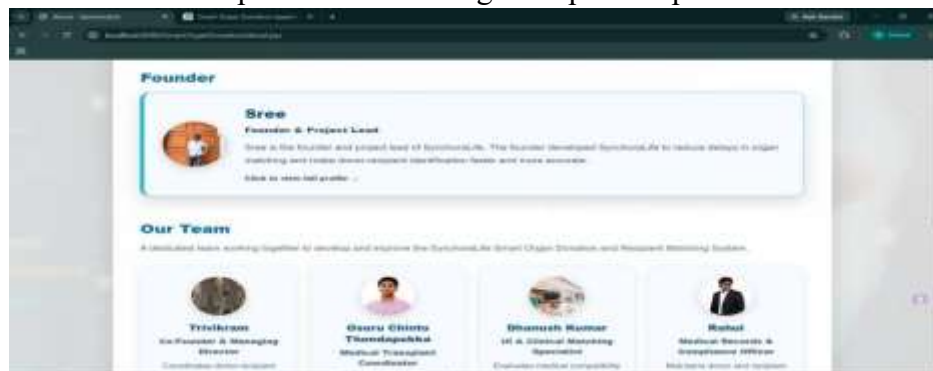


Figure 4: Screenshot of About Page

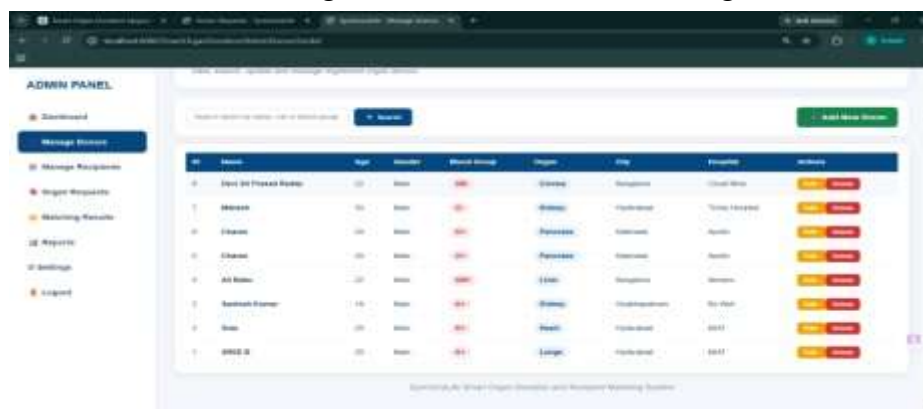


Figure 5: Screenshot of Admin Dashboard

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

- This project addressed the challenge of efficiently matching organ donors with suitable recipients. A smart organ donation system was developed using graph-based algorithms to represent and analyze donor-recipient relationships.
- The system successfully implemented donor and recipient registration, medical profile management, compatibility analysis, priority scoring, organ request management, and administrative monitoring. Testing confirmed that the major functions operated successfully.
- The proposed graph-based approach provides an organized method for analyzing multiple donor-recipient relationships and compatibility factors. Overall, the project demonstrates the potential of graph algorithms to improve organ matching and support efficient organ allocation while maintaining medical professionals' involvement in final decisions.

5.2 Future Work

- Integrate with real-time hospital databases.
- Include additional medical compatibility factors.
- Apply machine learning for intelligent match prediction.
- Develop a mobile application.
- Implement real-time notifications for donors and recipients.
- Improve security using advanced encryption techniques.
- Test the system with larger datasets.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Graph data structures and graph algorithms.
- Donor-recipient compatibility analysis.
- Priority-based allocation techniques.
- Java web application development.
- JSP and Servlet programming.

- MySQL database management.
 - Software testing and debugging. Engineering & Professional Skills Developed
 - Problem-solving and analytical thinking.
 - Software system design.
 - Database design and management.
 - Teamwork and task distribution.
 - Testing and debugging.
 - Technical documentation.
 - Project planning and time management.
- The most difficult challenge was designing the donor and recipient registration system with proper data management. This was solved by designing structured database tables and validating user inputs. A key decision was to separate donor and recipient information for better organization. I independently learned more about Java Servlets and database connectivity. If the project were repeated, I would improve data validation and security features.
 - The major challenge was analyzing multiple compatibility factors during donor-recipient matching. This was addressed by developing a structured compatibility scoring method. A key engineering decision was assigning different importance to matching factors based on system requirements. I independently learned about graph algorithms and their applications. In a future redesign, I would include more advanced medical compatibility parameters.
 - The most difficult task was implementing organ request, approval, and allocation management. The problem was solved by creating a structured workflow for request status updates. A key decision was to use priority-based allocation instead of simple sequential allocation. I gained additional knowledge about database transactions and status management. If repeated, I would improve real-time notifications and workflow automation.
 - The main challenge was integrating all modules and ensuring that they worked together correctly. This was solved through systematic testing and debugging. A key engineering decision was to develop a centralized administration and monitoring system. I independently learned about system testing and web application deployment. If the project were repeated, I would improve system scalability and reporting features.

REFERENCES

The project references should follow a consistent **IEEE citation style**. The following categories of resources used in the project should be properly acknowledged.

Literature

- [1] Relevant research articles and journal papers on organ transplantation, donor-recipient matching, and organ allocation algorithms.
- [2] Research publications on graph algorithms and their applications in healthcare systems.

Standards

- [3] International Organization for Standardization, *ISO/IEC 27001: Information Security Management Systems*.
- [4] OWASP Foundation, *OWASP Top 10 Web Application Security Risks*.

Websites and Technical Documentation

- [5] [Oracle, Java Documentation](#), accessed for Java programming and development.
- [6] [Apache Tomcat Documentation](#), accessed for web application deployment.
- [7] [MySQL Documentation](#), accessed for database development.

Software and Open-Source Resources

- [8] [Eclipse Foundation, Eclipse IDE](#), used for software development.
- [9] [Apache Software Foundation, Apache Tomcat](#), used as the web application server.
- [10] [Oracle, MySQL](#), used for database management.

Datasets

- [11] Sample and manually generated donor-recipient datasets used for system development and testing.

AI-Assisted Resources

- [12] [OpenAI, ChatGPT](#), used for assistance with documentation, content organization, and code explanation.

APPENDICES

Appendix A – Detailed Calculations

This appendix contains detailed calculations used in the donor-recipient matching and priority scoring process.

- Compatibility score calculation.
- Blood group compatibility scoring.
- Organ type matching.
- Medical urgency scoring.
- Recipient priority calculation.
- Final match score calculation.

Appendix B – Engineering Drawings / Schematics

This appendix contains the technical diagrams used to represent the proposed system.

- Overall system architecture diagram.
- System workflow diagram.
- Donor-recipient graph model.
- Database Entity Relationship (ER) diagram.
- Use case diagram.
- Matching process flowchart.

Appendix C – Source Code / Code Repository Information

This appendix contains essential source code excerpts from the project.

- Registration modules.
- Login and authentication.
- Donor and recipient management.
- Graph-based matching algorithm.
- Compatibility scoring.
- Priority allocation.
- Request and approval management.

The complete source code is maintained separately in the approved project repository or development environment.

Appendix D – Datasheets

This appendix includes relevant technical information used during development.

- Software and system requirements.
- Java and Apache Tomcat specifications.
- MySQL database requirements.
- Development environment details.

Appendix E – Experimental / Raw Data

This appendix contains sample data and testing information used for system validation.

- Sample donor data.
- Sample recipient data.
- Compatibility test cases.
- Priority scoring results.
- Matching results.
- System testing records.

Appendix F – Ethics / Institutional Approval Where applicable, this appendix includes:

- Ethical considerations related to medical data.
- User privacy and data protection measures.
- Institutional or project approval documents.
- Medical data usage limitations.

Appendix G – Project Meeting / Progress Records

This appendix contains evidence of project progress and team activities.

- Project meeting records.
- Weekly progress reports.
- Module development records.
- Faculty review feedback.
- Project milestone documentation.

Appendix H – User / Industry Feedback

Where applicable, this appendix contains feedback collected from:

- Project supervisors.

- Faculty members.
- Potential users.
- Healthcare professionals.

The feedback can be used to evaluate usability and identify possible improvements.

Appendix I – Publications / Patents / Competitions / Product Outcomes Where applicable, this appendix includes:

- Research publications.
- Conference presentations.
- Patent applications.
- Project competition participation.
- Product development outcomes.

Appendix J – Individual Contribution Evidence Mandatory for team projects.

This appendix provides evidence of each student's contribution to the project.

- Assigned modules and responsibilities.
- Source code contributions.
- Design and development work.
- Testing contributions.
- Documentation contributions.
- Screenshots or version-control evidence.
- Individual contribution percentage.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)